REVIEWER EVIDENCE + PRODUCT ACCURACY

IPO Intelligence: What the Prediction and Hold/Sell Scores Really Show

One report for two different jobs: Module 1 estimates whether an IPO opens above issue price; Module 2 gives post-listing hold/sell guidance. Their scores are not interchangeable.

MODULE 1 / IPO OPEN

0.810 AUC

189 later-year IPOs; 71.4% binary accuracy at the preselected 82% apply threshold.

MODULE 2 / HOLD-SELL

54.4%

250-IPO price-only, in-sample day+30 replay; always-sell baseline 54.0%.

1 The reviewer's Module 1 questions

The review asked for post-2024 testing, a confusion matrix or class-specific results, and probability calibration. The deployed model was fitted to 252 corporate IPOs listed in 2021-2024. The 2025 and 2026 records below are later-year retrospective tests, not a prospective trading trial.

Test data	IPOs	AUC	Brier	Accuracy	Always positive	Nonpositive recall
2025	108	0.785	0.172	70.4%	67.6%	85.7%
2026	81	0.843	0.160	72.8%	65.4%	85.7%
Combined	189	0.810	0.167	71.4%	66.7%	85.7%

At the 82% apply threshold, the combined test has 126 positive and 63 flat/discount openings. "Not apply" merges caution and elevated-risk calls; it does not mean the model predicted a guaranteed loss.

Combined 2025-2026 confusion matrix

Actual opening	Called APPLY	Called not-apply	Total
Above issue price	81	45	126
Flat or below	9	54	63

Plain-English read: 54 of 63 flat/discount IPOs were flagged not-apply (85.7% recall), but 45 profitable IPOs were also held back. Among 90 APPLY calls, 81 opened above issue price (90.0% precision). Accuracy beats the always-positive baseline by 4.7 percentage points, not by a dramatic margin.

MODULE 1 / PROBABILITY QUALITY

Do the stated probabilities resemble what happened?

AUC measures ranking, not whether a "70%" score wins about 70% of the time. The calibration check below groups the 189 later-year scores into broad ranges.

Predicted chance band	IPOs	Mean model score	Actually opened above issue
20-40%	41	35.8%	34.1%
40-60%	30	48.7%	46.7%
60-80%	28	69.5%	60.7%
80-100%	90	91.5%	90.0%

Summary: Model Brier score 0.1671 (lower is better), versus 0.2285 for a constant probability equal to the 2021-2024 training positive rate. Mean score 69.4% versus 66.7% actual positives. The 60-80% band is more optimistic than observed, and small groups limit precision.

Important boundary

A stratified IPO-level bootstrap puts later-year AUC roughly between 0.747 and 0.869; nonpositive recall roughly between 76.2% and 93.7%. These intervals describe uncertainty in this replay sample, not future market certainty. Historical final subscription books were assembled retrospectively and their public-availability times are not independently verified. Final books can arrive after a retail application cut-off, so this is not proof of a usable pre-deadline IPO application prediction.

2 Module 2: what was actually tested

Module 2 is a deterministic rule engine using current price, issue price, sector-relative performance, trailing highs and anchor-lock-in dates; news and disclosures can change a live verdict. It does not output a calibrated probability. We tested its price-only core as a dated historical decision, not the live news/disclosure experience.

Replay rule	Precise definition
Decision time	The close of the 31st available trading row (index 30) after listing. Only earlier/current rows were passed to the monitor.
Action mapping	SELL / REDUCE and WATCH / TAKE PROFIT count as sell-to-cash; HOLD and ACCUMULATE count as stay invested.
Outcome	Did the stock close above that decision-day close after 1 session or after 30 sessions? Flat is nonpositive.
Eligibility	250 corporate IPOs, 2021-2024. Excluded two trusts and 2 histories whose first local price was over 7 days late.
Unavailable inputs	No historical news or disclosure replay; disclosures=[] and news fetching disabled. Gross returns exclude fees, tax, spread and slippage.

MODULE 2 / OUTCOME CHECK

Hold/Sell monitor: next day and next 30 sessions

One decision per IPO avoids counting thousands of overlapping daily calls as independent tests. The same sell/stay decision is compared with two future horizons.

Outcome horizon	Right calls	Accuracy	Always sell	Always hold
Next trading day	122	48.8%	51.6%	48.4%
Next 30 sessions	136	54.4%	54.0%	46.0%

A right sell call means the later close was flat or lower. A right stay call means it was higher. This is directional classification, not a profit estimate.

What went right and wrong

Actual future return	Sell call	Stay call
Next day: flat or down	63	66
Next day: up	62	59
30 sessions: flat or down	73	62
30 sessions: up	52	63

Honest verdict: Next-day accuracy 48.8% is below the 51.6% always-sell baseline. The 30-session score 54.4% is only 0.4 point above always-sell. On this evidence, Module 2 has not demonstrated a useful predictive edge.

Separate, delayed-execution return check

A signal formed from session 30's close can first trade at the next session's close in this daily-price archive. From that execution close through 30 more sessions, selling to cash on a sell call and holding on a stay call returned +1.57% per IPO on average, versus +1.49% for holding every IPO. The difference is only +0.08 percentage point; its IPO-resampled 95% interval is -1.81 to +1.82 points. This omits trading costs and does not simulate next-open execution or real filings.

Why this is not a production accuracy claim

The rules contain comments describing a historical 254-IPO sweep, so this 250-IPO replay is not an independent holdout. The bootstrap 95% interval for the 30-session accuracy advantage over always-sell is -8.4 to +9.2 percentage points; it includes zero. The next-day interval is -11.6 to +6.0 points. The local price archive covers 254 older IPOs but none of the 189 2025-2026 IPOs, so a new-company later-year Module 2 test is not possible from this archive.

What must happen before quoting Module 2 accuracy publicly

Freeze rules and decision horizon; acquire dated 2025-2026 post-listing prices and as-published news/filings; record each decision before its outcome; evaluate first-day and 30-session actions separately against always-hold, always-sell and simple trailing-stop baselines; include costs, drawdown, missing-data rate and uncertainty. Do not use Module 1 AUC as Module 2 accuracy.

Evidence: matched source-file SHA-256 hashes; exact logistic replay checked against the deployed bundle; dated monitor replay from local prices. Project files: ml/model_bundle.json, ml/model_pipeline.py, core/monitor_engine.py and data/post_listing_prices.csv. The optional XGBoost comparator was not rerun here.